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International Mathematics Assessments for Schools

2023 /2024 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2023 / 2024 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. What is the sum of all the positive divisors of 2023?
（A）432　（B）2024　（C）2320　（D）2455　（E）2456

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2. Alice buys 9 souvenirs, each at a price of $1, $3, $5 or $25. She pays the cashier with several $10 bills. Which of the following is a change that she is impossible to get?
（A）$0　（B）$3　（C）$5　（D）$7　（E）$9

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3. In a certain school, there are less than 1000 students. The principal wants to align all of the students in the schoolyard for a flag ceremony. If the students are arranged in rows, 31 students per row, then there will be 13 students remaining. If the students are arranged 43 per row, then there will be 40 students remaining. How many students are exactly there in the school?
（A）900　（B）924　（C）931　（D）943　（E）986

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4. Two candles have different lengths and thickness, but each candle burns at a steady rate. The short candle and the long candle will burn out completely in 280 minutes and 210 minutes, respectively. Both candles were lit at the same time and their lengths were equal after 2 hours. If the initial length of the short candle is 8.4 cm, then what is the initial length, in cm, of the long candle?
（A）21 cm　（B）18.2 cm　（C）16.8 cm　（D）12.6 cm　（E）11.2 cm

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5. Four of the angles of a convex polygon are equal to 120° each. What is the minimum possible interior angle, in degrees, of all such polygons?
（A）30°　（B）60°　（C）90°　（D）120°　（E）Uncertain

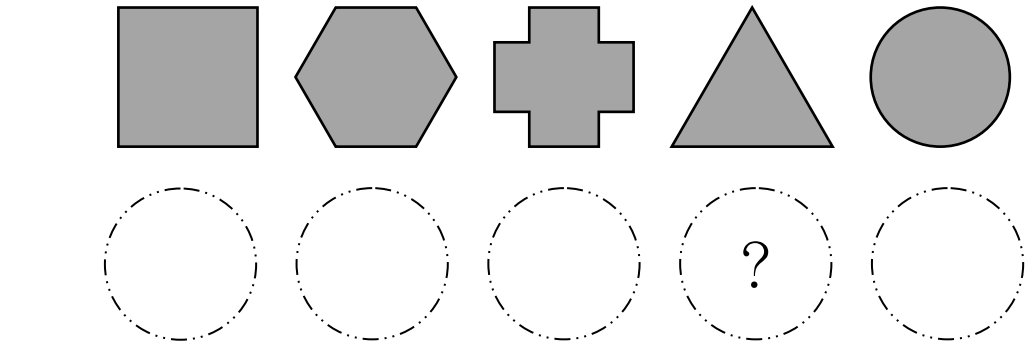
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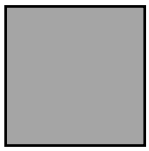
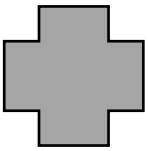
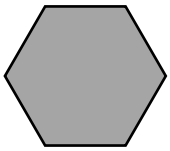
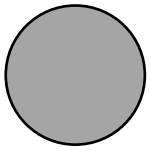
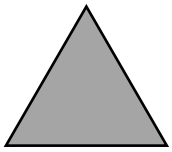
6. An urn contains n balls, which are numbered from 1 to n ($n \geq 3$). After a ball is drawn from this urn, the average of the numbers written on the remaining balls in the urn is 31.4. What number was written on the drawn ball?
（A）6　（B）7　（C）8　（D）10　（E）11

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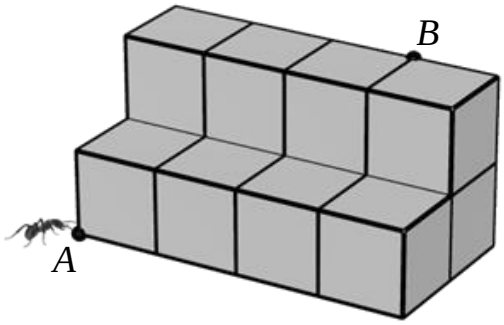
7. The lower row of shapes, which are all hidden, is a different sequence than the top row. The hidden row does, however, satisfy the following conditions:
   - Neither the cross nor the circle is adjacent to the hexagon.
   - Neither the cross nor the circle is adjacent to the triangle.
   - Neither the circle nor the hexagon is adjacent to the square.
   - The triangle is just to the right of the square.

   What shape lies in the 4th position from left to right in the lower row?



(A) 
(B) 
(C) 
(D) 
(E) 

8. The diagram below shows a stack of 12 equal cubes of side-length a m each. In each cube, the length of a diagonal of a face is b m. An ant is travelling on the surface of theses cubes from point A to point B. What is the length, in m, of the shortest path the ant can take?

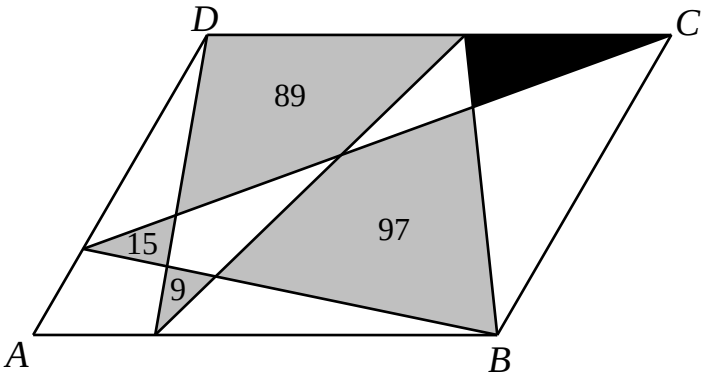


(A) $5a$ (B) $4b$ (C) $3b+a$ (D) $7a$ (E) $2b+3a$

9. The product of two positive integers, m and n, is 48. Let x be the number of distinct values of $m+n$ and y be the smallest value for $m+n$. What is the value of $x+y$?

(A) 15 (B) 16 (C) 17 (D) 18 (E) 19

10. In the diagram below, parallelogram *ABCD* is divided into some parts where only the shaded regions have known areas, in cm^2. What is the area, in cm^2, of the black triangle?



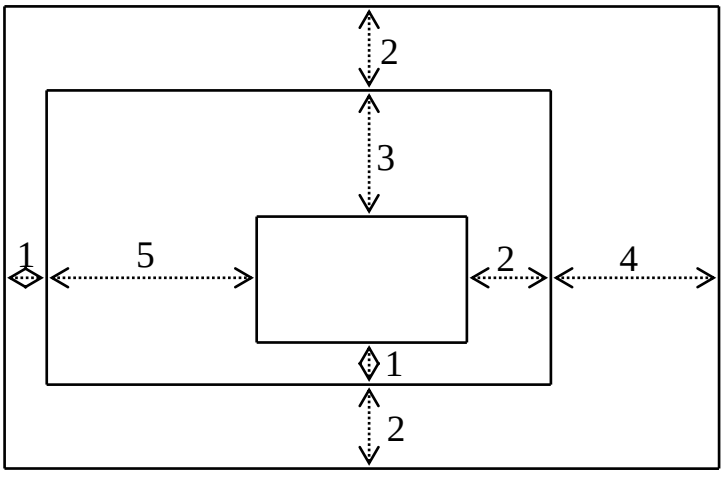
（A）12　（B）13　（C）14　（D）15　（E）16

Questions 11-20, 4 marks each

11. The average of fifteen distinct positive integers is 23. What is the maximum possible value for the second largest of these integers?

（A）126　（B）127　（C）128　（D）129　（E）130

12. As shown in the diagram below, all three rectangles have integer lengths and widths, in cm. If the area of the largest rectangle is 187 cm^2, then what is the area, in cm^2, of the second largest rectangle?



（A）15　（B）77　（C）84　（D）91　（E）96

13. The number 8000027 has exactly three prime factors. What is the sum of these three prime factors?

（A）39445　（B）39446　（C）39595　（D）39612　（E）39613

14. What is the last digit of the sum $1\times2+2\times3+3\times4+...+2022\times2023$?

（A）5　（B）6　（C）7　（D）8　（E）9

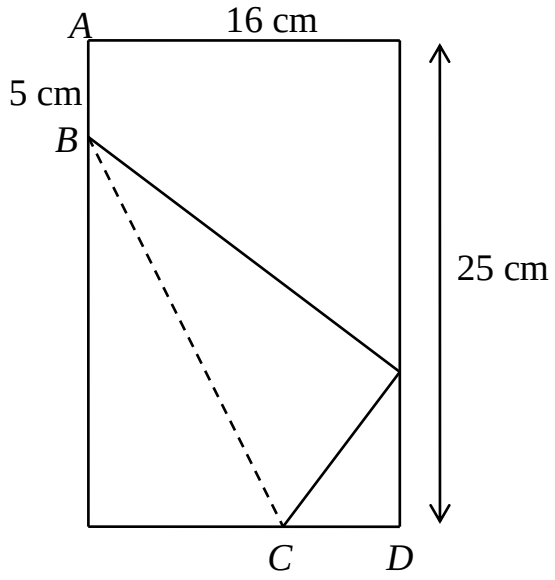
15. There are exactly \$500 in my bank account. Each time, I may either withdraw \$300 from my account, or deposit \$198 into my account from money previously withdrawn. What is the minimum amount of money that can be left in my bank account?

（A）\$0　（B）\$2　（C）\$8　（D）\$14　（E）\$20

16. Let $S(n)$ be the digit sum of a positive integer. What is the sum of all possible integers n for which $n+S(n)=2023$?

（A）4010　（B）4011　（C）4012　（D）4013　（E）4014

17. A $16\,\text{cm}\times25\,\text{cm}$ rectangle piece of paper is folded so that a corner touches the opposite side as shown in the diagram below. If $AB=5\,\text{cm}$, then what is the length, in cm, of CD?



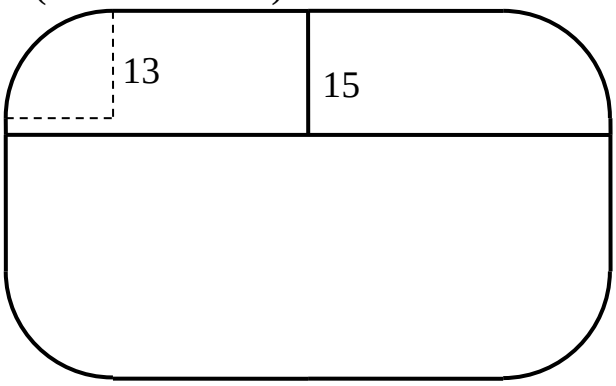
（A）3　（B）4　（C）5　（D）6　（E）7

18. There are four positive numbers a, b, c and d so that $a<b<c<d$. Which of the following inequalities is **INCORRECT**?

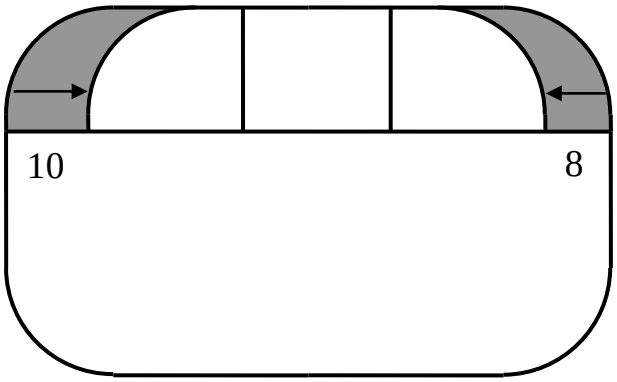
（A）$\frac{1}{a}+\frac{1}{b}>\frac{1}{c}+\frac{1}{d}$　（B）$\frac{d}{a}>\frac{a}{d}$　（C）$b+\frac{1}{c}<c+\frac{1}{b}$

（D）$\frac{1}{cd}>\frac{1}{b^2}$　（E）$c+\frac{c}{b}>b+\frac{a}{d}$

19. The diagram below on the left shows the closed window of an old bus. All curves forming the round corners are arcs of a quarter circle with radius 13 cm. The two sliding windows on top have equal size and their height is 15 cm. To open the window, the upper parts of the bus window are slid left or right, as shown in the diagram below on the right.
If the left window is slid 10 cm to the right, and the right window is slid 8 cm to the left, what is the total area, in cm^2, of the opening (the total shaded area)? (Take $\pi = 3.14$)



a closed window



an open window

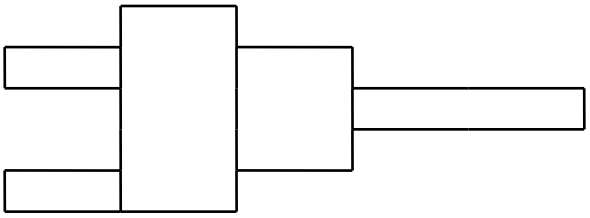
（A）234　（B）301.33　（C）270　（D）314　（E）uncertain

20. A basket contains some number of balls, each of them being painted with at least one of the two colours: green and red. It was observed that $\frac{3}{5}$ of the balls that are painted with green are also painted with red, and $\frac{2}{7}$ of the balls that are painted with red are also painted with green. What percent of the balls in that basket are painted in both colours?

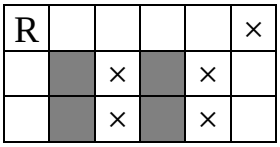
（A）16%　（B）24%　（C）35%　（D）40%　（E）56%

Questions 21-25, 6 marks each

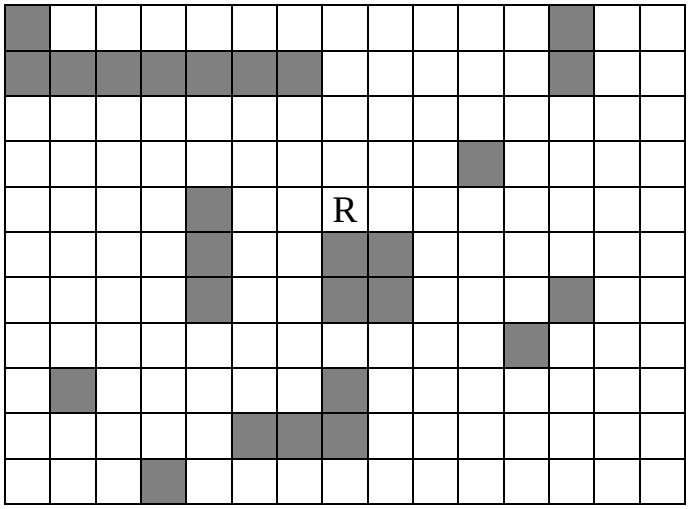
21. Alex designs a logo, which is divided into 5 regions. He wants to colour each region with one of the three colours: red, green or gold. No two regions with a common segment can have the same colour and each colour must be used at least once. How many possible colour schemes are available to Alex for this logo?



22. A robot can move up to 4 squares in one step. A step may not turn a corner. The robot cannot move to or cross any shaded square. For example, in the diagram below, R is the location of the robot at the start. By the rules, the squares marked with an × would take the robot two steps to reach.



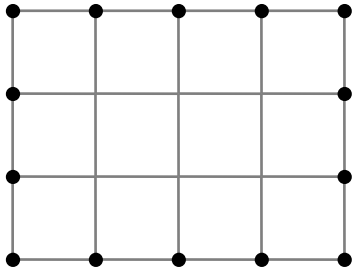
In the diagram below, what is the number of white squares that the robot **CAN NOT** reach within 3 steps? (Don't include the square containing R.)



23. A music band consists of 5 musicians. For each of the n songs the band played, only four musicians performed on stage, while the fifth musician rested. At the end of the concert, each musician counted how many times they performed on stage and they got 5 consecutive integers. What is the smallest possible value of n?

24. Let a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8 and a_9 be distinct non-zero digits so that $a_1 < a_2 < a_3 < a_4 < a_5$, $a_1 < a_6$, $a_2 < a_7$, $a_3 < a_8$ and $a_4 < a_9$.
How many different ways are there to assign the values to a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8 and a_9?

25. In how many ways can we construct an isosceles triangle by choosing three dots from the diagram below?



* * *